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Utilizing data driven methods to identify gender bias in LinkedIn profiles

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ABSTRACT

The growing use of Artificial Intelligence-enabled recruitment systems has become an important component of modern talent recruitment, particularly through social networks such as LinkedIn and Facebook. However, data overflow embedded in recruitment systems, based on Natural Language Processing (NLP) methods, may result in unconscious gender bias. The purpose of this work is to utilize a set of methods to analyze and detect textual bias in different groups. We analyzed a training dataset of fourteen thousand LinkedIn profiles, which was provided by a company named Talenya, and included job-candidates that fit IT-related positions. We aimed to detect textual self-presentation gender gap patterns, utilizing Term Frequency - Inverse Document Frequency (TF-IDF), word2vec and the Universal Sentence Encoder (USE) to code the data, and applied the kernel two-sample test for the purpose of determining whether men's and women's LinkedIn profiles have the same distribution. Additionally, we focused on quantifying and identifying repetition in skills representation in the LinkedIn profile by applying the TF-IDF and cosine similarity tools and compared the repetitiveness pattern of men's and women's profiles. Gender-based analysis was also carried out on smaller, more homogeneous groups of candidates, who share the same position type, geographical location and organizational seniority level. Finally, we provide theoretical and practical implications.

1. Introduction

Artificial Intelligence enabled recruitment has become the major tool for locating and ranking potential candidates for technology positions (Bondielli & Marcelloni, 2021; Hunkenschroer & Luetge, 2022). Gender gaps in the way men and women present themselves on LinkedIn are causing gender bias in Artificial Intelligence (AI) recruitment-enabled tools by placing fewer women in high-ranked places in lists of potential candidates. This was illustrated in 2018, when Amazon's experimental AI recruitment tool ranked job candidates in a range from one to five stars. Amazon's system taught itself that men software developer candidates were preferable and penalized resumes that included the word "female" (Dastin, 2018).

Evidence from the industry shows that AI-enabled recruiting do not always produce unbiased decisions about job applicants. Biases may penetrate into the datasets and effect the decisions of the AI algorithms, which may in practice strengthen minority and gender gaps (Soleimani, Intezari, & Pauleen, 2022). In order to develop less biased AI-enabled recruiting, a collaboration between Human Resources (HR) managers and AI developers for training algorithms and exploring algorithmic biases is vital. Drage and Mackereth (2022) deals with two claims in relation to the development and deployment of AI-powered HR tools: (1) recruitment AI can objectively assess candidates by removing gender and race from their systems, and (2) this removal of gender and race will

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make recruitment fairer, help customers attain their diversity, equality, and inclusion goals, and lay the foundations for a truly meritocratic culture to thrive within an organization. The authors argue that these claims are misleading. AI has the potential to improve recruitment practices, but it is important to recognize the limitations and potential biases of these systems. Collaboration between HR managers and AI developers is essential to ensure that AI-enabled recruiting is ethical, unbiased, and effective in achieving the above goals.

Gender bias in AI recruitment systems may have various origins, such as: a low-diversity talent pool; biased or inconsistent learning data along with unintended modeling procedures; gender gaps in self-presentation (Baumeister & Hutton, 1987; Foschi & Valenzuela, 2008); social and gender role perceptions (Eagly, Wood, & Diekman, 2000), or distribution of the value of attributes in a specific group different from the baseline distribution that is considered an unbiased group (Peng et al., 2019; Shahriari & Shahriari, 2017; Srivastava & Rossi, 2018). Clearly, potential bias in the decision-making process may result in discrimination based on gender, race, religion, and national origin (Black & van Esch, 2020). One approach for improving systems that take decisions based on machine learning algorithms is to develop fair algorithms. In Makhlouf, Zhioua, and Palamidessi (2021), the authors suggest a decision support model to identify fairness-related characteristics and to fit a fairness notion that will be relevant to the task at hand.

Previous research highlighted that the use of AI in recruitment may lead to unequal access to employment opportunities (Raub, 2018). Further studies have shown that algorithm-based decision-making may discriminate against minorities despite a fair computing process (Pessach & Shmueli, 2021). It has been accepted that if a bias exists in the training data, the machine-learning model may maintain the bias in the predicted results (Pessach & Shmueli, 2021). Including more suitable women and ethnic candidates in the top job ranking will not only improve those specific candidates' odds of being accepted for the positions, but also will create a less biased data training ground for future ranking. Data-driven manipulations and algorithms may assist in the creation of a diverse top-ranked candidates list, thus assisting in maintaining a bias-free recruitment environment (Deshpande, Pan, & Foulds, 2020; Houser, 2019). Recent work on factors that influence new-generation candidates to engage with and complete digital, AI-enabled recruiting, recommends leveraging AI to actively adjust the wording to decrease gender bias and increase candidate diversity (Van Esch & Black, 2019).

1.1. Research objectives

This study is aimed to achieve three main objectives. First, to provide research methods for detecting gender bias caused by gender differences in textual self-presentation in LinkedIn. These methods support AI-enabled recruitment systems. Second, to present a robust measurement method for textual bias analysis in various candidates' populations, specifically, in technical positions on LinkedIn. Finally, to gain insights on gender bias by analyzing both general and specific populations, grouping candidates by socio-demographic and organizational features, such as position type, seniority level, and geographical location (Mehrabi, Morstatter, Saxena, Lerman, & Galstyan, 2021).

The goal of this study is to provide research methods for detecting gender bias caused by gender gaps in textual self-presentation in Artificial Intelligence (AI)-enabled recruitment recommendation systems based on social networks. Our proposed methods aim to support AI recruitment systems to reveal bias in the textual data and take corrective action to eventually suggest a more diverse list of potential candidates. We present a method of measuring textual bias between different populations. Specifically, we aim to reveal gender bias associated with LinkedIn technical positions. The analysis is both general and specific, grouping the candidates by socio-demographic and organizational features, such as position type, seniority level, and geographical location (Mehrabi et al., 2021).

To explore whether two groups of texts are drawn from the same distribution, we propose a new approach based on encoding sentences into embedding vectors with Universal Sentence Encoder (USE) (Cer et al., 2018) as input data for a statistical test, the kernel two-sample test, which measures discrepancies between two multivariate probability density functions (Gretton, Borgwardt, Rasch, Schölkopf, & Smola, 2012). Our results reveal a statistically significant gender bias between men and women in most sub-groups.

When using algorithms for searching and ranking candidates, the number of repeated skills in the LinkedIn profile may impact on candidates' ranking. Therefore, in order to analyze the bias between men and women in the way skills are repeated, we use the classic approach of Term Frequency - Inverse Document Frequency (TF-IDF) with cosine similarity. The lists of similarity scores ranging between 0 to 1 of men and women are then compared via t-test to test the hypothesis that men and women do not (or do) repeat their skills in a similar manner.

In most of the t-tests we conducted, the results are biased against women, particularly in sub-groups that contain candidates from the fields of software development, Information Technology (IT), and product professionals. When grouping by location, the Silicon Valley area shows gender differences. When grouping by seniority level, there are gender differences at the low-to-mid management seniority level. In other sub-groups, the hypotheses of originating from the same distribution cannot be rejected; specifically, LinkedIn profiles of men and women who work in the field of data analysis, and also those of Quality Assurance (QA) professionals, are not significantly different. Men and women from New York City and Europe and those who are senior managers also do not show significant differences in textual LinkedIn presentation.

The rest of this paper is organized as follows. In Section 2, we review related literature related to gender difference on LinkedIn. The dataset and methodology are described in Section 3. Application of the methodology for identifying gender-based textual self-presentation differences in LinkedIn profiles is presented in Section 4. Results are shown in Section 5. Finally, Section 6 contains a discussion of this work, as well as its implications.

2. Related work and hypothesis development

2.1. Hypothesis development based on social and gender role perceptions

Our study delves and is supported by known socio-psychological theories. Since unconscious bias in self-representation refers to the tendency of individuals to present themselves in a way that aligns with their desired perception by others, this phenomenon may be influenced by various factors. For example, social norms, personal beliefs, and the desire for social acceptance (Jacobs, Berscheid, & Walster, 1971).

Social psychology offers the self-presentation theory, which suggests that individuals actively manage their self-image and the impressions they make on others (Baumeister & Hutton, 1987). The theory suggests that people strategically select and present information about themselves to create a favorable impression, which may be driven by unconscious biases. In social networks generally, and in LinkedIn more specifically, individuals often have control over how they present themselves through profile pictures (Tifferet & Vilnai-Yavetz, 2018), wording descriptions, and content they share. Thus, on LinkedIn individuals may selectively highlight specific career nodes, as well as downplay or omit others, with the intention of creating a specific impression. This can include showcasing achievements, experiences, relationships, and affiliations that align with socially desirable traits or identities.

Similarly, the Social Role Theory is a framework used in social psychology to explain the development of gender roles and the impact of societal expectations on individuals' behavior. The theory suggests that social roles, which are the sets of norms and expectations associated with particular positions in a given society, play a crucial role in shaping people's behavior, attitudes, and beliefs (Eagly et al., 2000). According to the theory, when individuals occupy a specific social role, for example: gender, they are expected to fulfill certain responsibilities, display specific behaviors, and adhere to particular norms and values associated with that role. These roles are often learned and reinforced through socialization processes (e.g. family, peers, social media). The theory proclaims that gender differences in behavior and attributes arise primarily due to the social roles assigned to men and women rather than inherent biological differences. For example, traditional gender roles may prescribe that men are expected to be assertive, independent, and competitive, while women are expected to be nurturing, caring, and cooperative. These role expectations can influence individuals' self-perception, aspirations, occupational choices, and social interactions. The theory also suggests that gender roles are not fixed but can change over time. Following these theoretical foundations, we propose that:

Proposition 1. *Self representation on LinkedIn and social roles perceptions may contribute to the development of bias based on stereotypes, thus reinforce inequalities due to distorted representation of individuals' experiences.*

2.2. Gender gaps in AI recruitment systems

The main data source used in AI-enabled recruitment is the professional network platform LinkedIn (Koch, Gerber, & De Klerk, 2018). In 2023, LinkedIn reports that there are more than 850 million members.[1] LinkedIn allows people to connect with and present themselves directly to colleagues and recruiters by sharing their photo, resume, skills, endorsements, and self-presentations via an Executive Summary (Zide, Elman, & Shahani-Denning, 2014). Research conducted recently on gender gaps among LinkedIn users across countries, ages, industries, and seniority measured women's participation in the workforce as reflected through their LinkedIn profiles and found that women are under-represented in technical professional fields such as software engineering and on websites such as GitHub, StackOverflow, or as Wikipedia editors, in a way that reinforces offline gender gaps (Kashyap & Verkroost, 2021). Kashyap and Verkroost (2021) suggest that it would be valuable to check gender differences in country-specific detailed analyses and to explore other detailed characteristics on LinkedIn, such as skills, which is one of the goals of this work.

Srivastava and Rossi (2018) noted that imbalanced gender distributions can result in AI bias in decision-support systems. While the issue of measuring and mitigating gender bias has been studied in recent years, the perpetuation of gender stereotypes through search engines and word embeddings has been established in psychological literature. To address this issue, Fabris, Purpura, Silvello, and Susto (2020) developed the Gender Stereotype Reinforcement (GSR) measure, which uses gender-related information encoded in word embeddings to quantify the extent to which a search engine reinforces gender stereotypes. Castelnovo, Cosentini, Malandri, Mercorio, and Mezzanzanica (2022) introduced a flexible, discrimination-aware decision-tree approach that enhances decision-tree classifiers to provide transparent and fair rules to end-users, addressing the increasing demand for transparency and fairness in AI-based decision-making systems. In addition, Şenel et al. (2022) proposed the Bidirectional Imparting (BiImp) method, which encodes gender-opposite concepts in a single dimension, to reduce gender bias in word embeddings. BiImp has the potential to mitigate biases and stereotypes in word embeddings and offers flexibility in studying them based on task or domain.

Several studies focus on LinkedIn gender gaps in visual self-presentation. For example, Tifferet and Vilnai-Yavetz (2018) analyzed differences in LinkedIn portraits used for self-presentation. They state that women are more likely to signal emotions while men are more likely to signal status. Sczesny and Kaufmann (2018) focused on differences between men's and women's LinkedIn pictures. Men cropped their photos to show greater facial prominence compared to women. One of the proposed methods to eliminate LinkedIn bias is by using a blind hiring process, eliminating names and pictures from LinkedIn profiles when they are presented to job recruiters (Rothman, 2018). Altenburger, De, Frazier, Avteniev, and Hamilton (2017) analyzed MBA graduates' LinkedIn profiles. The study results indicate that female users tend to have less content in the Summary and Job Description fields compared

[1] https://www.linkedin.com/company/linkedin

to male users. Figueroa (2017) studied discriminant learning for guessing demographic factors like gender, age, and location. He succeeded in building an NLP and machine learning model which determines a person's gender based on semantic analysis of age, industry, and second-level question categories. If these are inaccessible, the model infers them, to some extent, from textual sources. In Erfani, Hickey, and Cui (2023), the authors tackle a new research question associated with gender. These are gender stereotypes. They show, by using text mining analysis over LinkedIn data, that when a women reaches a leadership role, which is considered masculine, she faces the likeability vs. competency dilemma. Between these two poles, women do show a professional level that is equal to men, but are less likeable.

Recent studies analyze AI bias in recruitment systems utilizing various methodological tools. Some Natural Language Processing (NLP) methods, such as term frequency-inverse document frequency (TF-IDF) and machine learning tools have been utilized for investigating digital AI-enabled recruitment processes, reducing bias, and improving fairness (Roy, Chowdhary, & Bhatia, 2020). Stein (2018) use NLP, in particular a TF-IDF-based model, to predict hiring based on corporate cultural compatibility: work preferences, lifestyles, and ideology. In other work, TF-IDF was used to mitigate demographic bias in AI-enabled resume filtering, showing how a resume's wording differs by ethnicity. Deshpande et al. (2020) suggested a fair TF-IDF method to reduce bias and improve fairness in resumes matching job descriptions, mitigating socio-linguistic bias in job recommendations. Cosine similarity was applied for comparing a job description to a candidate's resume (Deshpande et al., 2020) and for comparing potential candidates and hired employees to predict the chances of a candidate fitting in to the organizational culture (Stein, 2018).

In LinkedIn profiles, keywords are usually embedded in the Skills section (Bastian et al., 2014) and the Executive Summary section (Bremner & Phung, 2015). The Skills section contains a list of abilities which serve as important keywords (for example: "SQL," "social media marketing," "software development" etc.). The Executive Summary is a short section that describes candidates' experience, areas of expertise, skills, and other competencies (Chalutz Ben-Gal, Avrahami, Pessach, Singer, & Ben Gal, 2021). However, it has been shown that different groups tend to present themselves differently on LinkedIn. Typically, profiles created by members of minority groups are shorter and include fewer keywords (Deshpande et al., 2020), compared to those created by white males, which is the majority group for technical positions for both employees and candidates. Thus, developers of automatic recruitment algorithms should first learn those differences and then incorporate bias correction in their algorithms. Talenya, a company that develops AI-enabled recruitment, analyzed over ten million LinkedIn profiles and reported that "women and ethnic minorities tend to underestimate themselves and present themselves less glamorously in a way that may hurt their ranking". Talenya also demonstrated that when candidates self-describe their skills and achievements on their LinkedIn profiles, female potential candidates write 34% less text compared to male potential candidates.

Two recent studies utilized LinkedIn to examine women's career paths and career choices. Career differences between genders were investigated in Hickey, Erfani, and Cui (2022), using features extracted from LinkedIn profiles. These features included the number of firms and positions and the person's network size, to name some. Machine learning classification algorithms were applied to the extracted features, and it was shown that the relationship between gender and career path can be learned with a high accuracy. Moreover, the study found that while men can develop a successful career and advance to leadership in a single company, women need to change work places in order to promote. Gender gaps in the entrepreneurship technology industry were detected in Miric, Yin, and Fehder (2023), by analyzing a large number of LinkedIn profiles. It was found that the gender gap in the technology entrepreneurship is larger than in the non-technology entrepreneurship industries. This gap may be explained by the fact that less women leave their technology jobs in order to fund new companies, and, that there is a higher number of men who are series entrepreneurs.

3. Data and methods

This work utilizes several complementary methods and tools for identifying and analyzing textual self-presentation gender bias in LinkedIn profiles. Fig. 1 provides a road map for the analysis that was carried out. The left side of the figure indicates the two textual sections that were extracted and analyzed from the LinkedIn profiles. These are the Executive Summary section and the list of skills in the Skills section. Our goal, again, is to identify self-presentation differences between men and women, based on these candidates' texts. The general approach is to represent the text components as vectors and then use this representation as input for statistical tests. For vector representation, we examined the Universal Sentence Encoder (USE) and TF-IDF (middle part of Fig. 1). Then, statistical tests such as t-tests and the kernel two-sample test (right side of Fig. 1) were carried out to identify gender bias. In the following, we describe the research setting, the dataset, and the methods.

3.1. Research setting

The dataset utilized for this study was extracted from an AI-enabled recruitment database, composed of millions of LinkedIn profiles. The dataset was provided by Talenya Inc., a company that has developed a new Talent AI application for recruitment. It is based on an AI data-driven algorithm that gives the recruiter a real picture of potential candidates and talent pools. Talenya's algorithms process LinkedIn data, they also integrate online data from other sources (such as GitHub). Talenya's patent feature for bias reduction adds additional fabricated skills to potential candidates' LinkedIn profile information. These are derived skills crafted by a machine learning algorithm that learns the skills of similar potential candidates. A machine learning algorithm, which is also tuned for bias reduction, matches a prioritized list of highly-matched potential candidates to recruiters' job descriptions. Finally, the candidate's ranking is calculated by Talenya's AI engine.

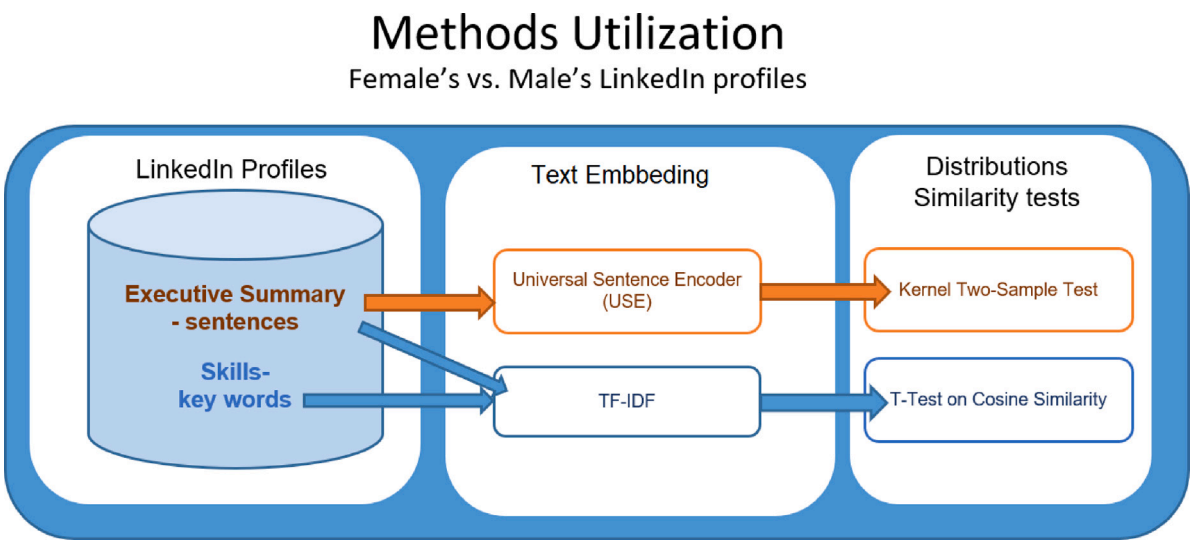


Fig. 1. Schematic description of the methods that were applied for textual analysis.

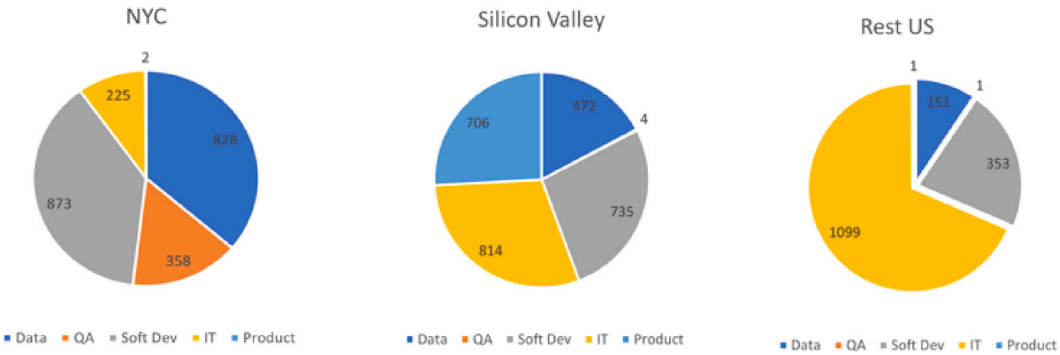


Fig. 2. Distribution of position types by US sub-regions. A large percentage of data analysts are located in NYC. We will show that this improves the similarity between men and women candidates in this area.

LinkedIn profiles do not include gender information. Talenya adds gender information with an algorithm that recognizes gender in the user profile for diversity and anti-bias purposes. Profiles that were not associated with gender information were removed from this study. We also removed all potential candidates who did not come from technical backgrounds (for example, accountants).

We used a dataset that was parsed by Talenya, and in one section included Talenya's patent feature for bias reduction in the analysis. We also used Talenya's algorithm for classifying the profile as one belonging to a male or female. Other than that, we based the analysis on raw, unprocessed data that was extracted from the profiles.

3.2. Dataset

The dataset used for this research was extracted from 13,687 LinkedIn profiles that are associated with 34 core technology positions. The extraction (parsing) process was performed by Talenya Inc., thus, we received the data in a table form. In this dataset, 2718 data points belong to female potential candidates (19.85%) and 10,968 belong to male potential candidates (80.15%), after unknown gender profiles are eliminated. This male–female distribution is similar to that currently found in today's technology industry.[2] The dataset's features are detailed in Appendix. Examples of the features are the number of positions to date, name of the current position, education level, years of experience, highest seniority, the largest company worked for, skills from different online sources, skills as listed on LinkedIn, the Executive Summary, current job description, highest seniority years of work, managerial years of work, average tenure and more. Fig. 2 displays descriptive statistics about the dataset in terms of socio-demographic factors.

[2] www.eeoc.gov/special-report/diversity-high-tech

3.3. Methods

In what follows, we describe four main methods that were utilized in this paper for text embedding and distribution similarity measures as described in Fig. 1 for sentences and words similarities.

3.3.1. Text embedding using universal sentence encoder

The Universal Sentence Encoder (USE) is a set of models for encoding sentences into embedding vectors that specifically target transfer learning to other NLP tasks (Cer et al., 2018). USE models are efficient and result in accurate performance on diverse transfer tasks. Transfer learning using sentence em- bedding tends to outperform word-level transfer when full sentences are transformed. The USE model we used is version 4 found in the TensorFlow Python library. This model has good performance, minimal amounts of supervised training data for a transfer task, and fast processing with limited computation resources. The model is freely available for download on Tensor Flow Hub. In this work, the Universal Sentence Encoder is applied for encoding sentences as a baseline for distribution similarity between men's and women's LinkedIn Executive Summaries.

3.3.2. Text embedding using TF-IDF

TF-IDF (term frequency-inverse document frequency) is a common technique for coding a set of documents. It is based on a numerical statistical measure that reflects how important a word is to a document in a collection or corpus of documents (Qaiser & Ali, 2018). Term frequency (TF) is a measure of the frequency of a term t that appears in a document d. Inverse document frequency (IDF) is a measure of the uniqueness of a term in the corpus d. Its role is to scale up the weights of rare terms and scale down the weights of common terms. In TF-IDF, each term is notionally assigned a different dimension, and a document is characterized by a vector. In this paper, TF-IDF is used to reveal patterns in textual representations that are associated with gender gaps. In particular, we generate two feature vectors that are based on TF-IDF for each candidate. The first vector is a TF-IDF representation of the candidate's Skills and the second vector is a TF-IDF representation of the candidate's Executive Summary, as seen in Fig. 1 (center box, bottom).

3.3.3. Text embedding using word2vec

Word2Vec (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013), a popular word embedding technique, represents words as numerical vectors in a multi-dimensional space. It captures semantic relationships between words by considering their context. The algorithm utilizes shallow neural networks in order to learn continuous word representations from large text corpora. Such a representation codes words to vectors, allowing to easily measure semantic similarities between words.

3.3.4. The kernel two-sample test

The kernel two-sample test is a framework for analyzing and comparing distributions of high-dimensional data (Gretton et al., 2012). It allows testing whether two sub-sets of data (here sub-sets that belong to women's and men's data points) are drawn from different distributions in a multi-dimensional space. A statistical test determines if the null hypothesis, which assumes that these samples are drawn from equal distributions, holds. Given that $X \sim p$ and $Y \sim q$, we test whether $p \neq q$. For simplicity, sets of equal sizes are considered. The statistical tests result in a p-value, which helps us to decide whether to accept $H_0 : p = q$ or the alternative hypothesis $H_A : p \neq q$. In this work, the kernel two-sample test is applied to compare the distributions of men's and women's LinkedIn features (as listed in Appendix). The aim of this test is to determine whether there exists gender bias in the numerical features that are drawn from LinkedIn profiles.

3.3.5. Cosine similarity

Cosine Similarity (CS) is a measure of similarity between two non-zero vectors, u and v, of an inner product space (Tata & Patel, 2007). CS is defined as the cosine of the angle θ between the vectors. When two vectors are in the same direction, the angle between them is $\theta = 0$ and $cos(0) = 1$. Two vectors that are orthogonal to each other have a similarity of 0. Cosine similarity can work on any number of dimensions, and it is most commonly used to measure distances in high-dimensional spaces. Cosine similarity is an intuitive measure of how similar two documents are in content and subject. Here, CS is used to measure the similarity between pairs of TF-IDF vectors. In particular, we will measure the similarity between the two TF-IDF vectors that code each candidate's information: their Skills and Executive Summary representations.

4. Identifying gender-based textual self-presentation differences in Linkedin profiles

The methods described in the previous section were applied for the detection and analysis of gender-related textual bias, in particular self-presentation differences in LinkedIn profiles. First, the entire population of men and women potential candidates in the dataset was examined by the methods described in Fig. 1. Then, we utilized the same methods across different socio-demographic and organizational sub-groups. This enabled us to extract HR analytics insights associated with self-presentation gender differences on LinkedIn across various geographic locations (e.g., Europe, US, sub-regions in the US), position or job types (e.g., software developers, QA, data analysts, product managers, and IT) and seniority levels (non-managers, low-mid and senior-top managers).

4.1. Detecting gender bias with the kernel two-sample test

In phase 1 of Fig. 1 (colored in orange), we performed two types of analysis with the kernel two-sample test. In the first, the kernel two-sample test was applied to the numeric features, as detailed in Appendix. The goal of this test is to determine whether the numeric features of men and women come from the same distribution. Textual information was not part of this test. Two datasets of numeric features were constructed, such that one holds features from men's profiles and the other one holds features from women's profiles. Both datasets were of the same size, the size being the number of women in the dataset. The test was applied five times for cross-validation. In each test, the dataset of men held different male candidates, while the dataset of women stayed constant (since it is smaller in size). The features were re-scaled to have zero mean and standard deviation of one.

Within this numeric-based feature analysis, the kernel two-sample test was repeated after replacing the "LinkedIn Skills Length" feature with the "All Skills Length" feature. The "All Skills Length" feature includes the Skills together with additional fabricated skills added by Talenya's machine learning algorithm. These skills were added to all potential candidates. The purpose of this step is to understand whether adding fabricated skills affects the distribution similarity of men and women potential candidates, thus influencing AI bias. Since the kernel two-sample test is a multi-dimensional test, and the modified field is just one of many explanatory variables, even a slight change in the results may indicate that this skills addition is influential.

The second analysis that was carried out in phase 1 combines the kernel two-sample test with the output of the Universal Sentence Encoder (USE). In order to compare the sentence distribution of men and women in LinkedIn's Executive Summary sections, we used the Universal Sentence Encoder for text embedding. We compared of a set of n women-encoded sections to the same size (n) of men-encoded sections. In order to accept or reject the hypothesis that men and women are drawn from the same distribution, we utilize the kernel two-sample test to identify bias in how men and women write their summaries across different populations, in different geographic locations, with different professions, and at different managerial and seniority levels. The test was applied to pairs of groups in both a cross-gender (men compared to women) and a within-gender manner (groups of men compared to other groups of men).

4.2. Detecting gender bias with TF-IDF, word2vec and Cosine similarity

In phase 2 (see bottom blue arrows in Fig. 1), TF-IDF was used for identifying bias between women and men via wording. As an initial step, to gain insight into the data, word clouds were applied to reveal the differences between the terms men and women potential candidates tend to use to describe themselves. To create the word clouds, stop words were removed, and TF-IDF vectors were computed for each of the candidate's Executive Summary text. Then, word cloud visualizations of men and women were plotted separately. These word clouds present the terms used with the size correlated to the frequency of the term used in the dataset (Nguyen, Chang, & Hui, 2011).

A second test aimed to quantify the repetition of Skills in the Executive Summary profile section. Each of these includes a parsed list of words. The parsing (word embedding) was done once with TF-IDF and once with word2vec. Since AI ranking mechanisms are somewhat similar to search engines, it is reasonable to assume that it is advantageous for candidates to repeat their skills in different sections of the LinkedIn profile. Therefore, when a candidate repeats the skills that are listed in the Skills section in the Executive Summary, the term achieves a higher significance, and the overall ranking of this candidate improves for a position that demands this skill as crucial or mandatory.

To quantify the skill repetition in the Executive Summary, TF-IDF or word2vec representations coupled with the cosine similarity distance were applied as follows. We separated the textual part of the dataset into two datasets, one consisting of male and the other consisting of female potential candidates. Each dataset was used as a corpus for the word embedding process. For each candidate, we had two sets of parsed words, one from the Skills and the other from the Executive Summary. Then, word embedding vectors were created from both Skills and Executive Summary features for each candidate. Last, the cosine similarity between the Skills and Executive Summary embedded representation vectors was calculated for each candidate. This yields a similarity score between 0 and 1 for each candidate, where 0 means that none of the Skills were repeated in the Executive Summary. A resulting value that is close to 1 means that more skills were repeated. To understand if men and women potential candidates repeat their skills similarly, we use a t-test to detect differences between the potential candidates' cosine similarity results. Algorithm 1 summarizes the above steps.

In order to gain detailed insights on self-presentation differences, we implemented Algorithm 1 on the full dataset, and then on category subgroups. These subgroups were defined according to the candidates' socio-demographic features, different position types, seniority levels, and locations.

5. Results

First, exploratory data analysis, which plots the correlation between the features in Appendix, excluding the textual string features, is presented in Fig. 3. In order to compute the correlation analysis categorical features (like Gender, Seniority Level and Current Job Have Description) were converted to integer values. It is apparent that there is a strong correlation between "Current Job Max Description Length" and "Executive Summary and Description Grade", meaning there is a positive clear relationship between the length of the job description and the total grade the profile receives on the quality of his/her executive summary and description. Furthermore, it can be seen that gender has a weak correlation with all features. The correlation results point to the fact that gender is not a linear predictor for lower or higher values of any other numeric feature. This directs us to investigate the textual features, focusing on embedding vectors which represents words and sentences, as a basis for exploring gender differences.

Next, we apply the methods described in the previous sections and in Fig. 1, to reveal gender textual presentation differences according to different socio-demographic and organizational sub-groups.

Algorithm 1 Comparing men vs. women LinkedIn Text Repetitions

Input:

- A list of men's LinkedIn skills $S^M = \{s_1^M, s_2^M, \ldots, s_m^M\}$.
- A list of men's executive summaries $E^M = \{e_1^M, e_2^M, \ldots, e_m^M\}$.
- A list of women's LinkedIn skills $S^F = \{s_1^F, s_2^F, \ldots, s_n^F\}$.
- A list of women's executive summaries $E^F = \{e_1^F, e_2^F, \ldots, e_n^F\}$.

Output: T-test p-value score for man and woman text repetition similarity.

1: Create a joint list for each gender $L^M = S^M \cup E^M$, $L^F = S^F \cup E^F$
2: Create TF-IDF or word2vec embedding vectors from the L^M corpus and the L^F corp
3: For each person i extract his/her two associated embedding vectors from L^M or L^F. Denote the skills vector by s_i and the executive summary vector by e_i.
4: For each pair s_i and e_i compute a cosine similarity, $c_i = \frac{s_i \cdot e_i}{\|s_i\| \|e_i\|}$ and store it in C^M or C^F, according to i's gender.
5: Preform a t-test between C^M and C^F.

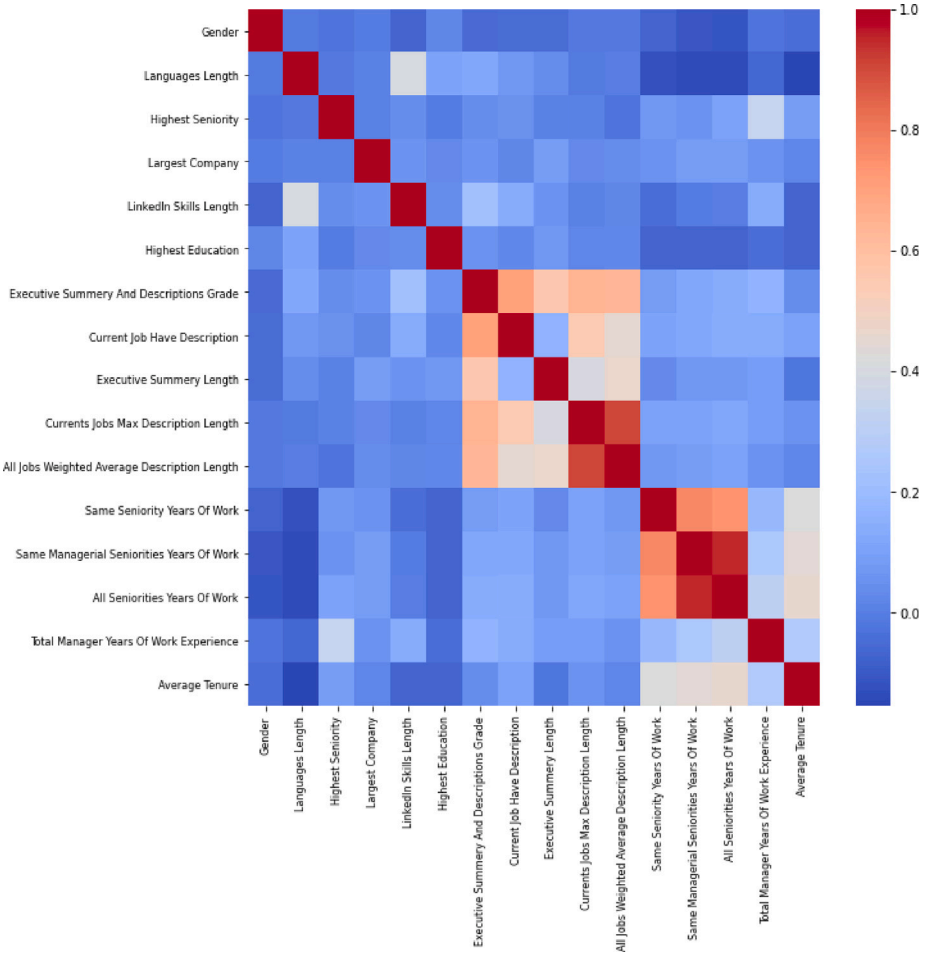


Fig. 3. Features correlation map. Red denotes a strong correlation and blue denotes a weak correlation. Gender has a weak correlation with all other numeric and categorical features. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Kernel two sample tests results on LinkedIn numerical data.

	LinkedIn Profiles	LinkedIn profiles + Artificial skills
Entire Dataset	0.179	0.183
Data Analysts	0.685	0.657
Software Developers	0.004*	0.03

*Bonferroni p-value < 0.025

5.1. The kernel two-sample test results

The goal of these tests was to compare men and women candidates based only on their numeric features, ignoring the textual data. Two tests were applied in a cross-validation manner on all profiles' numerical features, such as years of job experience, seniority level, and the number of LinkedIn skills (LinkedIn Skills Length).

The first test was applied to the original data that was parsed from the LinkedIn profiles. The results are plotted in the center columns of Table 1. In the second test, the input data includes a bias-correction feature from Talenya, which adds fabricated skills to the skills of each candidate. The additional fabricated skills come from a black-box algorithm of Talenya, which finds skills from other websites such as Github, from free text in the profile and from similar candidates, then adds them to the candidate in order to improve diversity. Thus, features that quantify the number of skills per candidate are influenced by this correction.

The results are presented in the left columns of Table 1. In the first test, which is presented in the top row, groups of equal size of men and women candidates were drawn from the entire dataset, at random. The distribution of the pairs of groups was compared. The test was repeated five times and the mean result is presented. In the middle and bottom rows, the kernel two-sample test was performed on sub-groups that were defined by position type: data analysts and software developers. We aimed to test whether the numerical feature distribution for men and women from these specific subgroups was drawn from the same distribution.

It can be seen in the first row, center column of Table 1, that when we compare the distributions of men vs. women in terms of numeric LinkedIn features from the entire dataset, there is no significant difference. When we repeat the test with the numerical features that include Talenya's additional skills, as seen in Table 1, first row, right column, the p-value slightly increases. This indicates that overall, the addition of the skills is relevant.

In Table 1, middle and last rows, the tests are applied to sub-groups that only hold candidates for data analyst and software developer positions. It can be seen in Table 1, middle and last rows, center column, that there is no significant difference between the numerical features of men and women data analysts. In addition, adding the fabricated skills to this sub-group did not improve the numeric-based similarity. In Table 1, bottom rows, center column, we notice the significant differences in numerical features between man and woman software developers. The addition of fabricated skills decreased the p-value (Table 1, bottom rows, right column). The results were computed using the Bonferroni correction to control for Type I error across the two tests. Bonferroni p value $= 0.05/2 = 0.025$.

5.2. Word clouds

The TF-IDF Word clouds are presented in Fig. 4. They emphasize the differences in wording between men and women potential candidates on LinkedIn profiles. Results indicate that men use more technical skills compared to women, while women tend to use soft skills. For example men use terms such as software engineer, technology, software development, application, and product, while women use softer skills such as management, business, and professional.

5.3. Textual self-presentation bias in the executive summary section

The Executive Summary sections of men and women were embedded with Universal Sentence Encoder and compared with the multidimensional kernel two-sample test (as presented in Fig. 1 top line). The kernel two-sample test was applied to homogeneous sub-groups from the data, examining both between- gender and within-gender differences.

On the left column of Table 2, we present the groups that were tested. The center column of Table 2 presents the between-gender results while the right column presents the results of the within-gender tests. It can be seen that for the within-gender groups, there were no significant differences, as expected, while for the between-gender comparisons, there are significant differences in the way men and women working as data analysts, software developers and IT professionals, present themselves.

5.4. Textual self-presentation bias in skills repetition

Textual self-presentation bias in skills repetition was examined in order to quantify how candidates repeat their own skills in the Executive Summary section. The methodology was described in Section 4, Algorithm 1. The algorithm results are comprised of the angle values between candidates' wording in these two LinkedIn sections. Performing a t-test on these two gender-related angle vectors shows a significant gender difference. We first applied Alg. 1 to the entire dataset, when the words were coded once

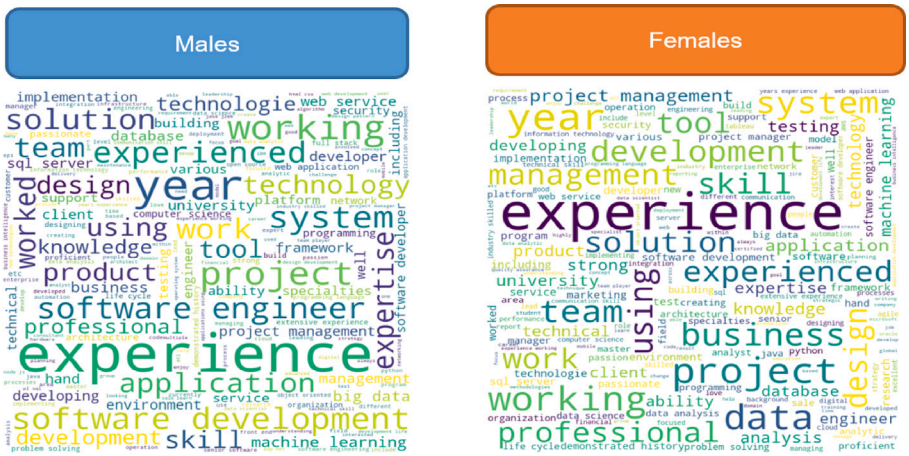


Fig. 4. Word cloud of men's and women's LinkedIn Executive Summary corpus. Men use more technically-oriented skills compared to women. Men use terms such as software engineer, technology, software development, application, and product, while women use softer skills such as management, business, and professional.

Table 2
Textual Self-Presentation Bias in Executive Summary.

Position type	*p*-value between-gender tests	*p*-value within-gender tests (men only)
Data Analyst	0.0001*	0.5393
Software Developer	0.0001*	0.6503
QA	0.5736	0.903
IT	0.0009*	0.7849
Product	0.0602	0.0947

*Bonferroni *p*-value < 0.01

Table 3
t-test text repetition results based on Alg. 1.

Sub-Group	p-value women vs. men
NYC	0.3439
Silicon Valley	1.67E−07*
Europe	0.343
Non-managers	0.08
Low-Mid Managers	0.0005*
Senior-Top managers	0.52
Data Analyst	0.535
Software Developer	6.79E−47*
QA	0.4174
IT	1.4E−74*
Product	4.31E−19*

*Bonferroni *p*-value < 0.0045

with TF-IDF and once with word2vec. In both cases, we found significant gender-differences related to the repetition of skills, with p-values of 0.0215 and 0.0004 for TF-IDF and word2vec, accordingly.

The test was additionally performed on different sub-groups from the data, defined by position type, location, and seniority level. In these tests, TF-IDF was utilized as an encoding method. Using the Bonferroni correction *p*-value (0.05/11 = 0.045), we see that the hypothesis of both genders originating from a similar distribution cannot be rejected for the data analyst and QA candidates. In Table 3, the t-test results of Algorithm 1 applied to the different subgroups are presented.

The results demonstrate differences by regions; these may be related to the job-type distribution that was presented in Fig. 2. In terms of seniority level, it seems that women in senior positions present themselves on LinkedIn in a similar way to men. Finally, women whose roles are those of data analysts, QA, and product management repeat their skills in the Executive Summary as men do. This is moderately consistent with the results plotted in Tables 1 and 2.

Table 4
Summary of results.

	Method	Test goal	Features	Population	Main findings	Reference
1.	Kernel two-sample test	Does it contribute to add artificial skills?	Numeric	All Data Data Analysts Software Developers	No sig. difference No sig. difference Significant	Table 1
2.	Word clouds	Are there gender-based wording differences?	Executive Summary Text	All Data	Men use technical words, women soft skills	Fig. 4
3.	USE & Kernel two-sample test	Do men's and women's textual self-presentation have the same distribution	Executive Summary Text coded by USE	Data Analysts Software Developers QA IT Product	Significant gender differences for data analysts, software developers and IT	Table 2
4.	TF-IDF, word2vec & cos. similarity	Comparing men vs. women text repetition	Embedded Skills and Executive Summary text	All Data	Significant gender differences	Section 5.4
5a.	TF-IDF & cos. similarity	Comparing men vs. women repetitions when grouped by location	Embedded Skills and Executive Summary text	NYC Silicon Valley Europe	Significant gender differences in the Silicon Valley	Table 3 top rows
5b.	TF-IDF & cos. similarity	Comparing men vs. women repetitions. when grouped by seniority level	Embedded Skills and Executive Summary text	Non-managers Mid-level Seniors	Significant gender differences for the Mid-level managers	Table 3 middle rows
5c.	TF-IDF & cos. similarity	Comparing men vs. women repetitions. when grouped by position type	Embedded Skills and Executive Summary text	Data analysts Software developer QA IT Product	Significant gender differences for software developers, IT and product	Table 3 bottom rows

5.5. Summary of results

The results of this work are summarized in Table 4. The table emphasizes the textual gender gaps that were inspected by a variety of tools. Gender differences are less pronounced when analyzing numerical data (see row 1), but are easily noticeable even by deriving a simple word cloud presentation (row 2). More sophisticated tests that are based on TF-IDF, word2vec and USE vectorized word representation followed by a quantitative test point out to significant textual gender differences in LinkedIn profiles (rows 3–4). Last, rows 5a–5c demonstrate the ability to detect sub-populations that may require extra care in terms of correcting actions by recruiters and AI enabled recruitment systems.

6. Discussion

Identifying and mitigating gender bias in professional and organizational settings is important for the purpose of promoting gender equality and diversity in the workplace (Kidron & Chalutz Ben-Gal, 2022; Pessach & Shmueli, 2022). LinkedIn is one of the most popular professional virtual platforms, generating immense data that can be used to identify gender bias in profiles and help promote a more inclusive professional environment. This study attempts to provide a novel approach focusing on a research gap associated with textual self-presentation habits, utilizing machine learning and advanced data analysis techniques. We introduce a new research method to explore existing bias associated with textual self-presentation, integrating kernel two-sample tests detecting difference based on multi-dimensional feature vectors (Srivastava & Rossi, 2018). We utilized statistical tests combined with textual TF-IDF analysis for quantifying text repetition. These tools may serve as a framework integrated with AI recruitment systems to identify biased subgroups. In order to rule out that bias exists in the numerical features, a kernel two-sample test was utilized to verify that men's and women's profiles come from the same distribution. Thus, we were able to conclude that the differences do not originate from their objective features (e.g., experience, education) but rather stem from their textual self presentation. These findings are in line with previous research showing that men dominate most skills in the technology industry (Haranko, Zagheni, Garimella, & Weber, 2018). Our results strengthen previous research on gender differences in LinkedIn's Executive Summary section (Altenburger et al., 2017), while providing an extended framework for pointing out the origin of these differences. The distribution of mens' and womens' Executive Summaries was measured by the Universal Sentence Encoder followed by the kernel two-sample test. The results clearly illustrate the gender bias in most of the position-type sub-groups of profiles (not including QA

and product profiles). Table 2 demonstrates the comparison between a group of women and men, versus a group of men with another group of men from the same position category and of the same size, emphasizes that men are drawn from the same distribution while women deviate considerably. Using a combination of TF-IDF, cosine similarity, and t-test, we identified gender self-presentation differences in skills repetition in the Executive Summary section and the use of technical language. In most cases, women potential candidates tend to less utilize text and technical skills wording in the Skills and Executive Summary sections. These results, as the previous tests, vary across subgroups and populations, although similar trends are maintained (Table 3). For example, women software developers, low-mid managers, and the Silicon Valley differ from men in skills repetition while data analysts, top-senior managers, and NYC origin profiles are similar across gender. We found that the influencing factors on textual self-presentation gender gaps depend on socio-demographic and organizational features (e.g. location, position type, seniority level). For instance, the gap seen for software developers is very much different than that seen for data analysts, where men and women come from the same distribution. Smaller textual gender gaps were found in locations such as Europe and NYC when compared to most of the U.S. areas. Similarly, textual gender gaps at senior levels are minor when compared to low level managers and non-managers sub-groups.

6.1. Theoretical and practical implications

Unconscious bias plays an important role in the self-presentation process because individuals may internalize societal stereotypes and norms, which can influence their perception of what is considered favorable or desirable. For example, someone may emphasize their professional accomplishments to be perceived as competent or highlight their social activities to be seen as outgoing and popular. These decisions may be influenced by unconscious bias about gender, race, age, or other social categories, as suggested in socio-psychological theories, such as the Social Role Theory. It may stem from a natural human desire for social acceptance and the subconscious influence of societal expectations. Nevertheless, these norms are be reflected in online media tools like LinkedIn, thus effect the outcome of automatic algorithms that process and make decisions based on this data.

Utilizing data-driven methods for identifying gender bias in LinkedIn profiles include various benefits. For example, achieving objectivity goals in employee recruitment (Pessach & Shmueli, 2022); Improved data scaling. Because of the vast amount of data associated with LinkedIn it allows for the identification of trends and patterns that may not be apparent through individual profile analysis (Sela & Ben-Gal, 2018). Furthermore, data-driven methods enable insights into the specific types of language and behaviors that may be contributing to gender bias in LinkedIn profiles. This can help to inform targeted interventions and strategies to promote gender equity and diversity.

Theoretical contributions of this study are twofold. First, we extend the contribution of Kashyap & Verkroost, who studied gender gaps in women's participation in different areas of the labor force (Kashyap & Verkroost, 2021), by analyzing gender gaps in textual self-presentation in the Skills and Executive Summary sections of LinkedIn. This study demonstrates valuable insights into the prevalence and nature of bias in professional settings. However, it is important to be aware of the limitations and ethical considerations associated with this approach, and to use the results to inform targeted interventions and strategies to promote gender equity and diversity in the workplace.

Second, based on previous literature, we propose that differences in gender-based self-presentation may originate from women's lower level of self-esteem compared to men, as repeatedly suggested in McMullin and Cairney (2004). Previous research found that even when women are equal to men in computing skills, they are less confident (Liberatore & Wagner, 2021), therefore may not mention their full skill set on their profile.

There are several practical implication to this study. First, providing hands-on tools for developers of AI systems in their aim to reduce gender bias in recruitment systems (Altenburger et al., 2017). Second, we provide recommendations which may assist organizations and decision-makers in reducing gender bias in AI-enabled recruitment tools. These recommendations are aimed at women candidates, HR professionals, and AI developers. HR professionals should be aware of gender differences relevant to specific socio-demographic data such as position type, geographical locations, and job seniority levels. Additionally, when forming the skills requirements for a job, recruiters should include technical as well as soft skills, as this is not equally distributed across gender (as seen in Fig. 4). Furthermore, we recommend that AI developers identify bias by studying pairs of groups' distributions and add skills adaptively by AI mechanism, according to the bias level that exists in the socio-demographic and organizational sub-group.

Finally, our recommendations for female job seekers are to be more aware of job requirements and relevant candidates' competitive profiles skills, include more relevant skills in the Skills section, and repeat and elaborate on them in the Executive Summary section of the LinkedIn Profile.

7. Limitations

This paper addresses the challenges of unbalanced data and data quality control in gender classification using LinkedIn profiles. The initial dataset exhibits a significant gender imbalance, with 80% men, which can lead to biased classification results. To mitigate this, we used techniques of under-sampling of the majority class and re-sampling of the minority class. Data quality control is another challenge, as unmaintained profiles compared to active ones can introduce noise and affect classification accuracy. Strategies such as data pre-processing, outlier detection, and quality control measures can help mitigate this issue. Addressing these challenges is crucial for fair and accurate gender classification in LinkedIn profiles. Last, our conclusions regarding gender differences rely on a dataset of 14 K LinkedIn profiles, therefore may be limited, and would benefit by being extended to a larger dataset.

CRediT authorship contribution statement

Vivian Simon: Conceptualization, Methodology, Software, Writing – original draft & final preparation. **Neta Rabin:** Conceptualization, Methodology, Supervision, Writing – original draft & final manuscript preparation. **Hila Chalutz-Ben Gal:** Conceptualization, Data Availability, Supervision, Writing – original draft & final manuscript preparation.

Data availability

The authors do not have permission to share data

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Appendix

Feature title	Description	Data type
Id	Talent ID in our database	Numeric string
Gender	Male/Female	Categorical
Position Id	Position ID in our database	Numeric string
Num. Of Current Positions	Num. of job positions until now	Integer
Current Positions	Current talent position	String
Education Length	The number of degrees	Integer
Languages Length	Number of spoken languages	Integer
Highest Seniority	Highest seniority level	Categorical
Largest Company	The size of the talent's largest company	Integer
LinkedIn Skills Length	Number of skills the talent listed	Integer
LinkedIn Skills	Skills text	String
Derived Skills Length	Number of derived skills by Talenya.inc	Integer
Derived Skills	Derived skills from the talent's resumes	String
Non-Derived Skills Length	Number of skills listed on the talent's resume	Integer
Non-Derived Skills	The skills list from the talent's resumes	String
Highest Education	Highest degree, level of education	Categorical Integer
Exc. Summary & Descriptions Grade	Overall score of profile text quality	Integer
Current Job Have Description	Does the current job have textual description?	Boolean
Executive Summary Length	Length of executive summary in number of words	Integer
Executive Summary	Executive summary text	String
Currents Jobs Max Description Length	Current position description in number of words	Integer
All Jobs Weighted Avg. Description Len.	Average num. of words in each job description	Integer
Industry	Industry name	String
Location – city	City name	String
Location – Country	Country name	String
Geo Location X	Geographic location's x axis	Float
Geo Location Y	Geographic location's y axis	Float

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